Graph Database Performance Benchmark Report

Benchmark comparison of Neo4j, Memgraph, CognoDB and FalkorDB

1. Executive Summary

The benchmark used a common graph dataset containing 49,683 nodes and 100,000 relationships. Six core query workloads were completed for the database comparison: Count Nodes, Point Lookup, 1-Hop Traversal, 2-Hop Traversal, 3-Hop Traversal and Aggregation. FalkorDB recorded the lowest measured average latency in all six core workloads in the completed runs.

The results are workload- and configuration-specific. They should therefore be interpreted as experimental measurements for this project rather than as a universal ranking of graph databases.

2. Dataset and Validation

Dataset: 49,683 nodes and 100,000 relationships.
Relationship type: KNOWS.
Node label: Person.
Validation: The completed database checks reported 49,683 nodes and 100,000 relationships for the imported benchmark datasets.

3. Core Query Benchmark Results

Workload	Neo4j	Memgraph	CognoDB	FalkorDB
1-Hop Traversal	105.16	334.48	341.80	29.49
2-Hop Traversal	114.04	478.35	393.70	29.92
3-Hop Traversal	261.29	907.97	877.11	31.80
Aggregation	123.70	367.90	353.20	24.16
Count Nodes	86.15	318.36	321.89	25.00
Point Lookup	105.40	325.83	330.80	28.92

Values are average latency in milliseconds; lower is better.

4. Detailed Latency Metrics

Neo4j

Workload	Average	Minimum	Maximum	P50	P95
Count Nodes	86.15	70.95	150.29	81.91	108.73
Point Lookup	105.40	76.65	502.33	100.07	117.25
1-Hop Traversal	105.16	88.86	242.30	101.11	130.42
2-Hop Traversal	114.04	89.92	268.86	107.98	151.15
3-Hop Traversal	261.29	184.35	728.71	243.57	406.17
Aggregation	123.70	106.54	198.06	121.31	148.15

Memgraph

Workload	Average	Minimum	Maximum	P50	P95
Count Nodes	318.36	288.62	456.04	314.82	359.52

Point Lookup	325.83	294.01	470.97	320.08	356.48
1-Hop Traversal	334.48	295.74	799.97	319.83	407.64
2-Hop Traversal	478.35	307.87	881.57	369.90	769.54
3-Hop Traversal	907.97	611.35	3548.30	838.19	1133.32
Aggregation	367.90	325.83	463.25	342.82	448.37

CognoDB

Workload	Average	Minimum	Maximum	P50	P95
Count Nodes	321.89	286.10	446.50	319.30	376.94
Point Lookup	330.80	295.06	477.54	319.96	399.77
1-Hop Traversal	341.80	292.08	1263.87	320.10	426.90
2-Hop Traversal	393.70	300.83	772.14	323.84	694.28
3-Hop Traversal	877.11	610.85	1356.78	945.91	1095.96
Aggregation	353.20	311.20	529.83	342.37	438.82

FalkorDB

Workload	Average	Minimum	Maximum	P50	P95
Count Nodes	25.00	22.92	31.82	23.54	27.98
Point Lookup	28.92	27.66	32.49	28.64	28.82
1-Hop Traversal	29.49	27.13	37.56	28.62	31.26
2-Hop Traversal	29.92	27.54	39.31	28.73	31.93
3-Hop Traversal	31.80	30.04	35.31	31.34	32.83
Aggregation	24.16	21.76	26.21	24.13	26.02

5. Additional FalkorDB Workloads

Indexed / Filtered Lookup: Average 151.59 ms; Minimum 121.32 ms; Maximum 249.63 ms; P50 129.52 ms; P95 194.47 ms. The Person.id range index was verified as operational with 49,683 indexed documents and zero indexing failures.

Mixed Read/Write: 5 concurrent clients, 200 total operations, total time 4.40 seconds, throughput 45.50 QPS, average latency 67.32 ms, P50 57.56 ms and P95 123.29 ms.

6. Key Findings

• FalkorDB had the lowest average latency for all six completed core workloads.
• FalkorDB's core-query averages ranged from 24.16 ms to 31.80 ms.
• Neo4j was the second-fastest database in the supplied core benchmark results.
• Memgraph and CognoDB showed substantially higher average latency in the deeper 3-hop traversal workload.
• The 3-hop traversal workload produced the largest latency gap among the four databases.

7. Limitations

The ingest benchmark was attempted for FalkorDB but did not complete because of connection/time-out problems. Therefore, no ingest throughput or load-time value is included in the final comparison. The mixed read/write and indexed lookup tests were completed only for FalkorDB in the recorded results, so they should not be presented as four-database comparisons.

8. Conclusion

Under the benchmark configuration and workload used in this project, FalkorDB achieved the best measured latency across the six core query workloads. Neo4j provided the next-lowest measured latency, while Memgraph and CognoDB were slower in these particular runs. The findings demonstrate the importance of workload-specific benchmarking and should be interpreted together with the stated limitations.

9. Reproducibility

The benchmark artifacts include Python scripts for connection checks, data import, core query benchmarks, FalkorDB indexed lookup, mixed read/write testing and result collection. The consolidated CSV contains the measured latency values used in this report.